



Semiconductor Equipment
Electron Beam Lithography Systems

Solutions for Innovation

JBX Series



JEOL Ltd.

In order to increase the production efficiency, JEOL offers electron beam lithography systems for a variety of applications.

JEOL has supplied a complete line of systems for 50 years, since the development of its first Electron Beam Lithography (EBL) system in 1967.

Today, JEOL is the world's top supplier of EBL systems. Our customers' applications of the systems range from mask/reticle production for high-end devices and nano-lithography research to ultrafine, direct write exposures of communication devices in their factories.

JEOL will continue pursuing higher writing accuracy, higher throughput and easier operation in advancing the development of advanced EBL systems.



JBX-8100FS



JBX-9

*cy and development speed of devices,
stems optimized*



500FS



JBX-3200MV
& JBX-3050MV

JBX-8100FS

Electron Beam Lithography System

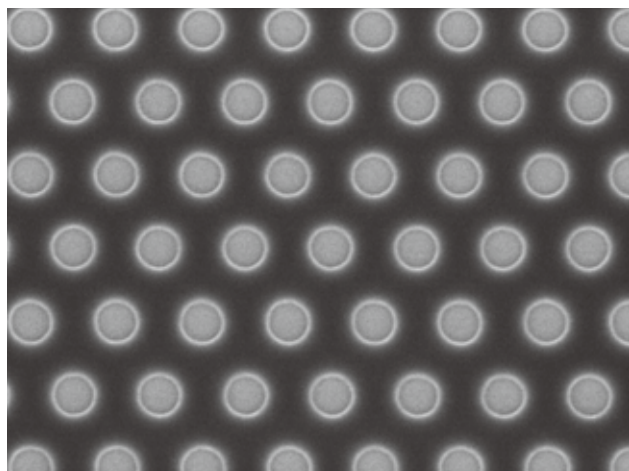
- Self contained and compact
- Extreme accuracy while maintaining high throughput
- Built with JEOL's renowned Electron Optics for ultimate stability
- Easy operation to obtain excellent results.
- Upgradeable and modular platform to enable you to improve the performance as your research progresses.

Principal Specifications

Writing method:	Spot beam, vector scan & step and repeat
Accelerating voltage :	100 kV/ 50 kV/ 25 kV
Electron beam emitter :	ZrO/W Schottky
Substrate size :	Maximum 200 mm wafer
Minimum line width :	≤8 nm @ field center
Field stitching accuracy :	±20 nm @ 1 mm field
Overlay accuracy :	±20 nm @ 1 mm field
Scan speed :	125 MHz



Photonic Pattern



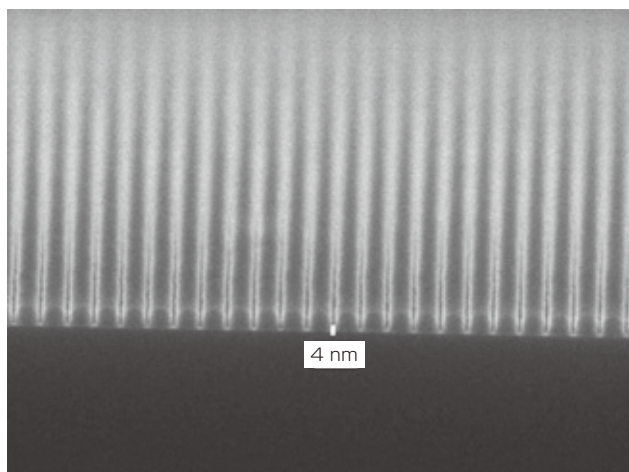
100 nm

Grating Pattern



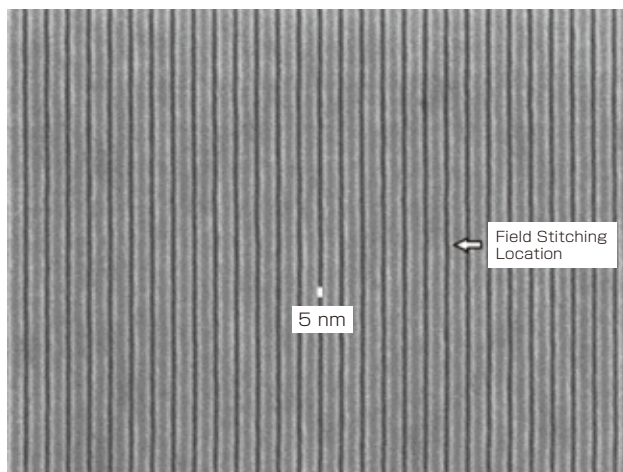
1 μm

4 nm Line Pattern



100 nm

Field Stitching Pattern



JBX-9500FS

High Throughput, High Precision Direct Writing EB Lithography System

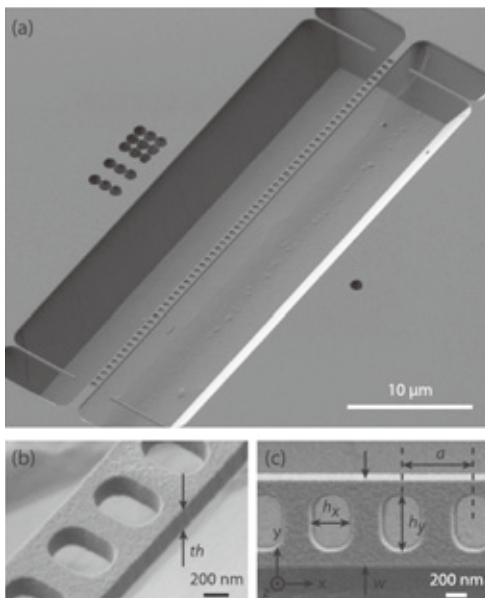
- State of the art, high throughput high precision spot beam system.
- 100 kV accelerating voltage
- High resolution Schottky emitter
- Up to 300 mm wafer loadable
- High performance field stitching and overlay accuracy
- Application from nano technology to leading edge device production

Principal Specifications

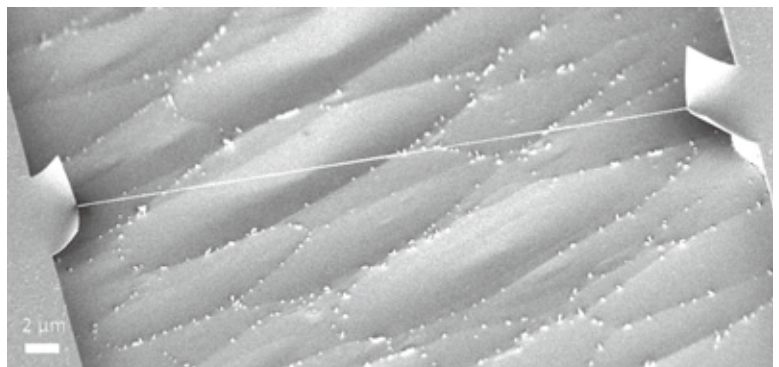
Writing method :	Spot beam, vector scan & step and repeat
Accelerating voltage :	100 kV
Electron beam emitter :	ZrO/W Schottky
Substrate size :	Maximum 300 mm wafer
Minimum line width :	≤ 10 nm @ 1 mm field
Field Stitching accuracy :	± 10 nm @ 1 mm field
Overlay accuracy :	± 11 nm @ 1 mm field
Scan speed :	120 MHz
Position accuracy within plate :	± 11 nm @ 60 mm



High-Q Silicon Carbide Photonic-Crystal Cavities. Jonathan Lee, et al., University of Rochester



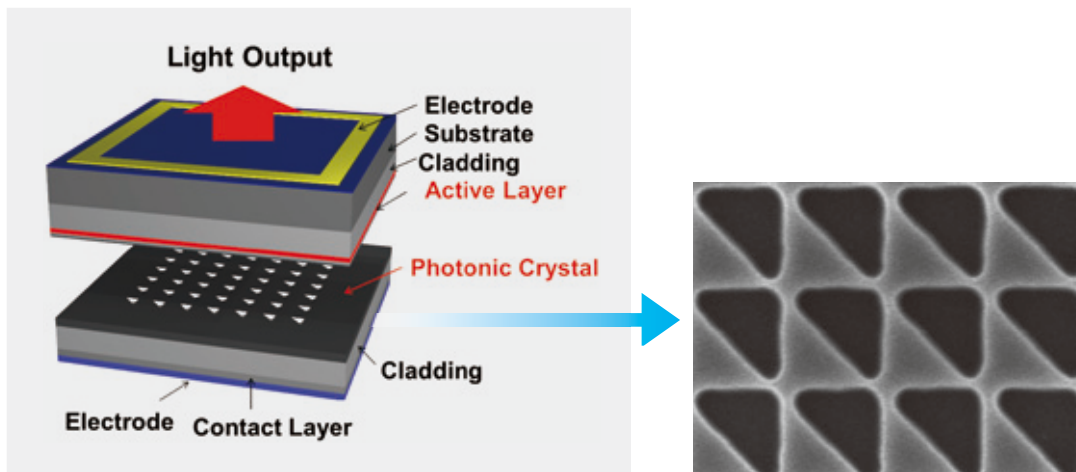
Ultra-thin Superconducting Film Coated Silicon Nitride Nanowire Resonators for Low-Temperature Applications. Abhilash Sebastian, Cornell University



Alan R. Bleier (Cornell NanoScale Facility) JEOL Electron Beam Lithography at Cornell

Application Examples

Photonic crystal laser



Courtesy of S. Noda (Kyoto University)
K. Hirose *et al.*, *Nature Photonics* **8**, 406-411 (2014)

SEM image

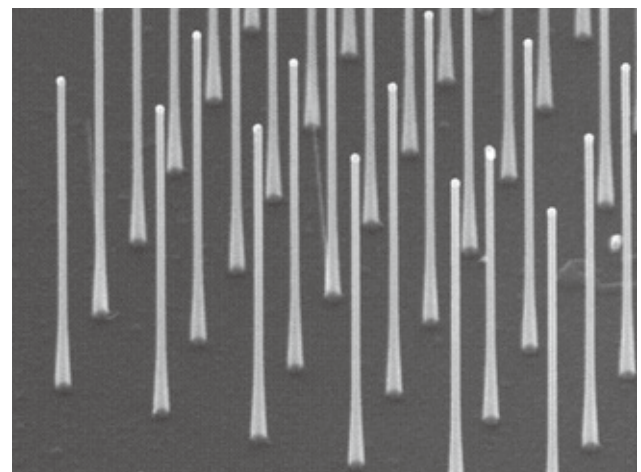
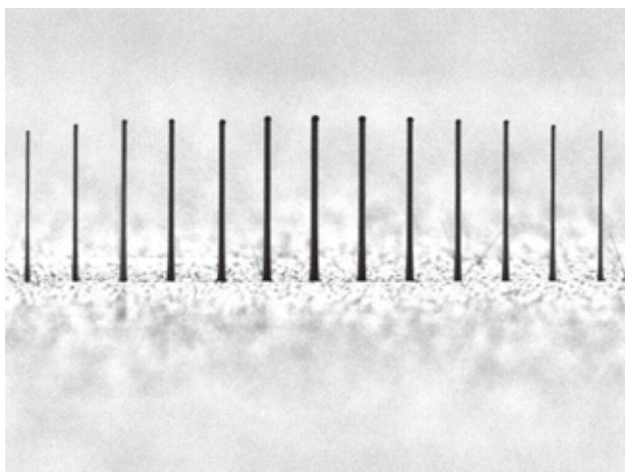
Selective-area growth of InGaAs nanowires on Si(111)



Courtesy of T. Fukui (Hokkaido University)
K. Tomioka *et al.*, *Nature* **488**, 189-192 (2012)

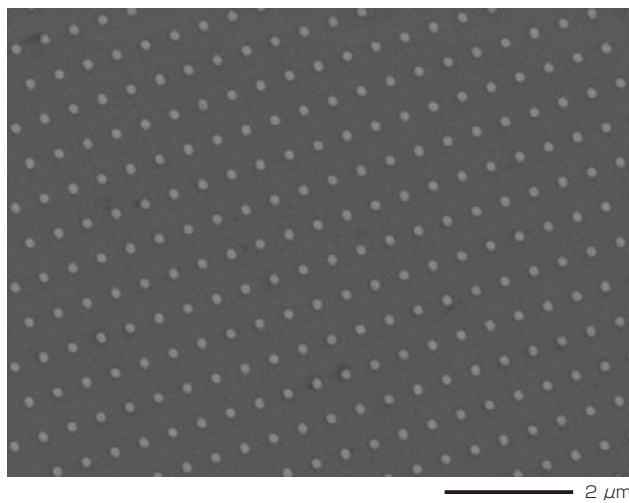
Scanning electron microscope (SEM) image showing selective-area growth of InGaAs nanowires on Si: vertically aligned nanowires were grown on a Si(111) substrate. The surface between the nanowires is covered with an SiO_2 film. The nanowires are 90 nm in diameter and 760 nm in height.

Ge nanowires grown from lithographically patterned Au dots

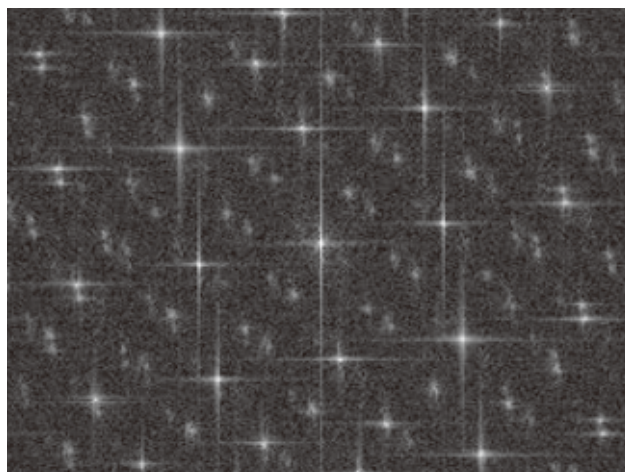


Courtesy of S. Dayeh (University of California, San Diego) and T. Picraux (Los Alamos National Laboratory).
S. Dayeh *et al.*, *Nano Lett.* **10**, 4032 (2010)

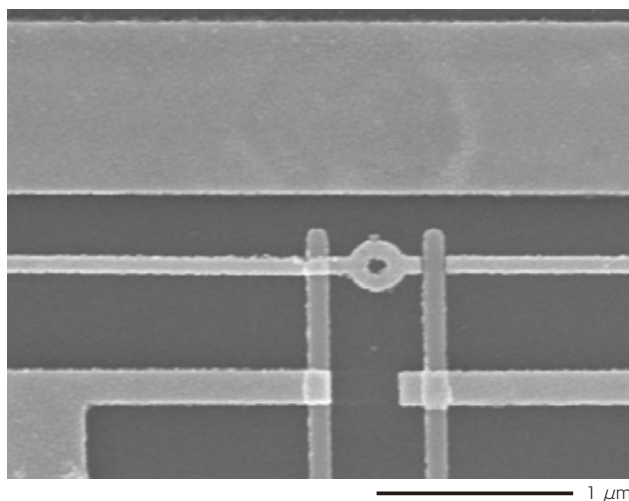
Exchange bias in large scale nanodot arrays



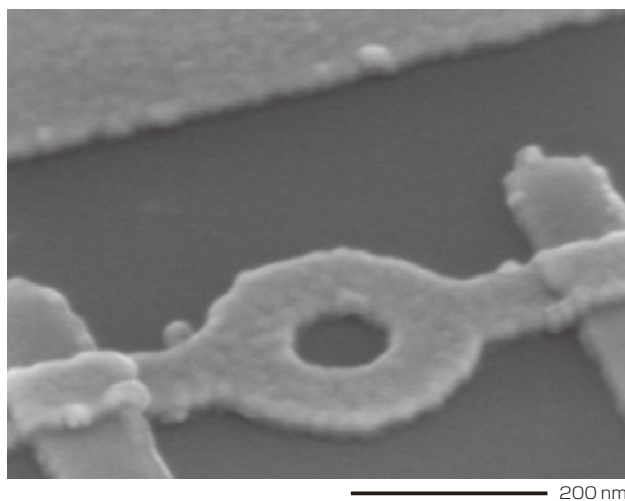
Courtesy of E. Linfield (University of Leeds) and A. Hirohata (University of York).



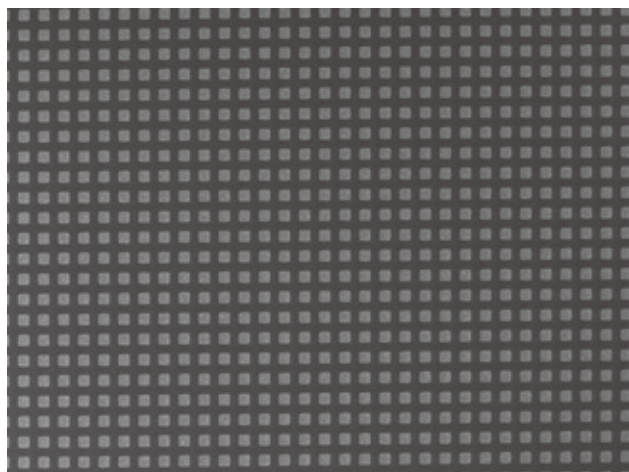
Non-magnetic nano-ring for spin-current operation



Courtesy of E. Linfield (University of Leeds) and A. Hirohata (University of York).



High current exposure



1 μm square pattern (100 kV, 100 nA).

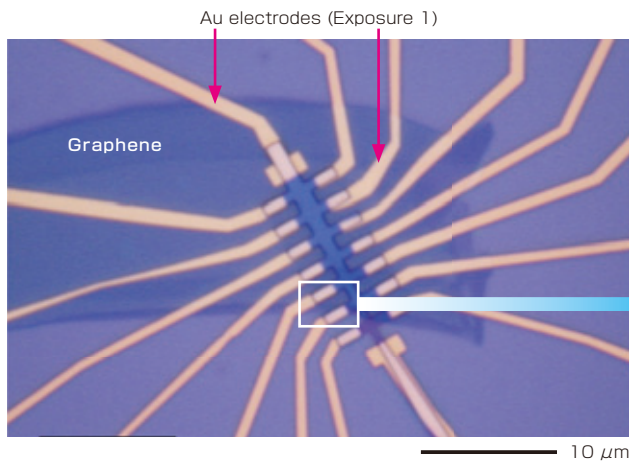


200 nm line pattern (100 kV, 100 nA).

Application Examples

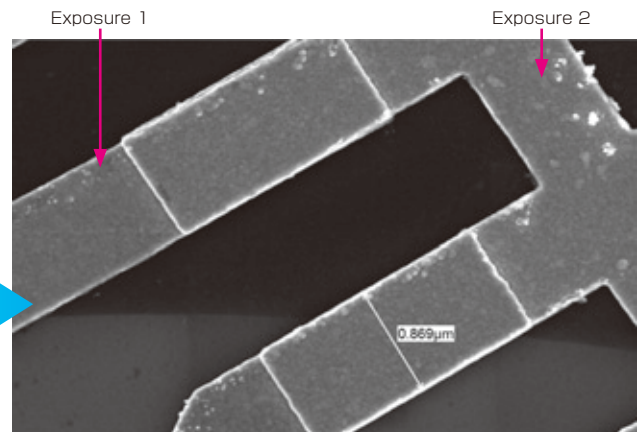
Electrodes patterning on a Graphene

Image of an optical microscope

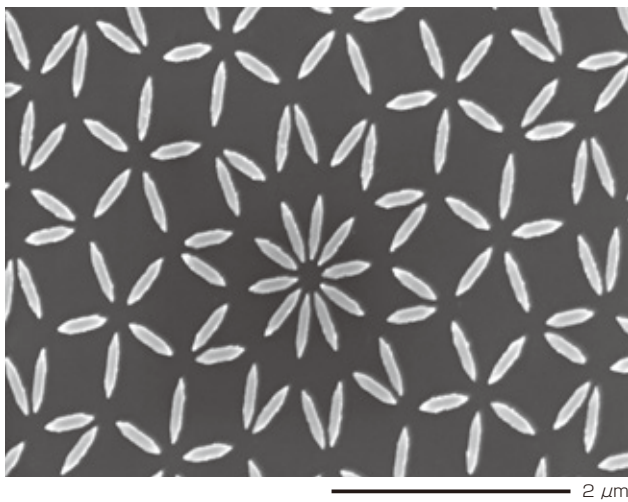


Courtesy of Graphene Research Centre, National University of Singapore.

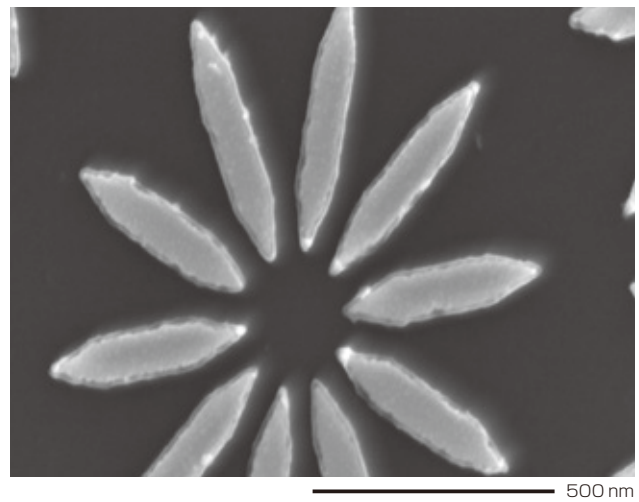
After lift-off Au at gate image of SEM



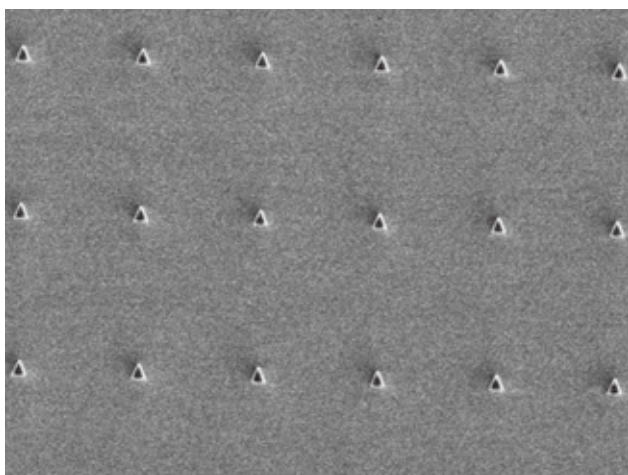
Penrose tile ice array of nanomagnet



Courtesy of J. Morgan, C. Marrows (University of Leeds), and A. Stein (Center for Functional Nanomaterials, Brookhaven National Laboratory).

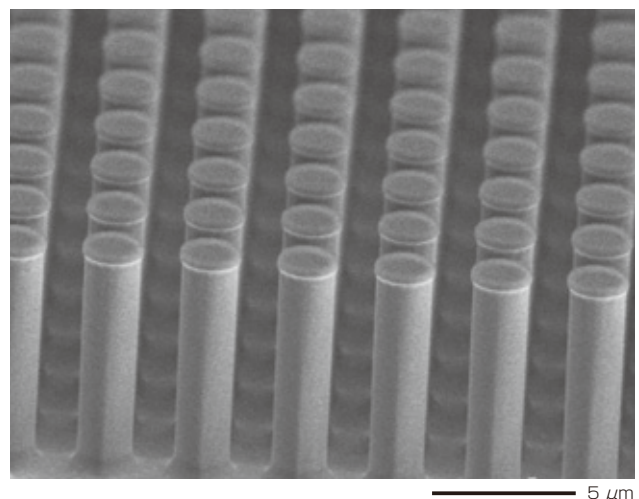


HSQ triangular pillars on diamond substrate



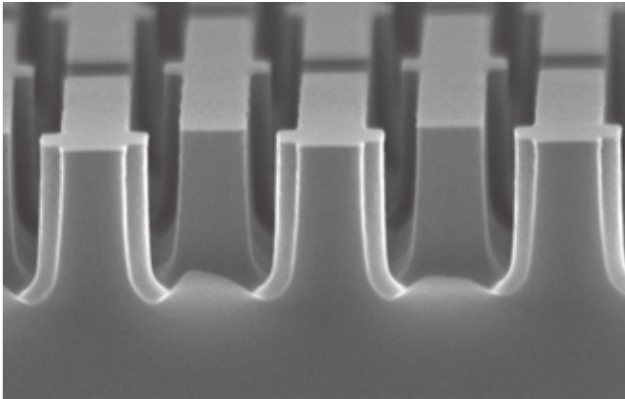
Courtesy of L. Li (Columbia University) and D. Englund (Massachusetts Institute of Technology).

Photonic crystal pattern etched in silicon



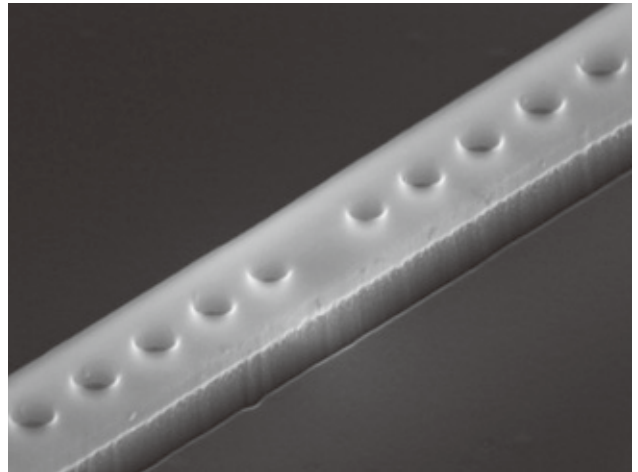
Courtesy of J. Beck and I. Kyriassis (Columbia University).

Etched Si mesa brick-wall pattern optimized for minimal RIE laginduced depth variation for photovoltaic application



Courtesy of D. Johnston (Center for Functional Nanomaterials, Brookhaven National Laboratory).

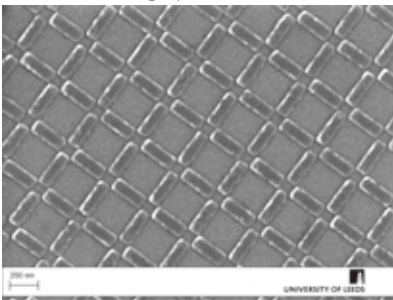
1-D photonic crystal silicon waveguide cavity



Courtesy of R. Grote, and R. Osgood (Columbia University).

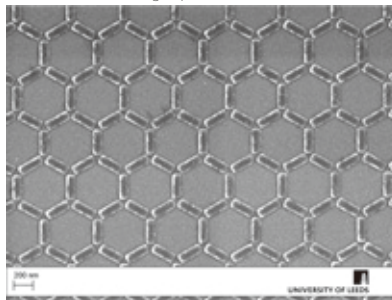
Spin ice array of nanomagnet

CoFeB square spin ice pattern Si_3N_4 membrane



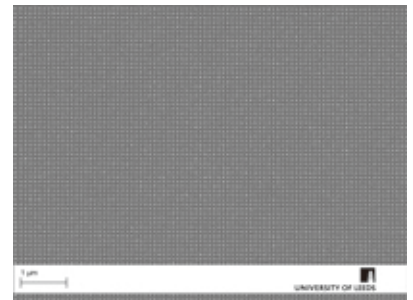
Courtesy of M. Rosamond and S. Morley (University of Leeds).

CoFeB Kagome spin ice pattern Si_3N_4 membrane



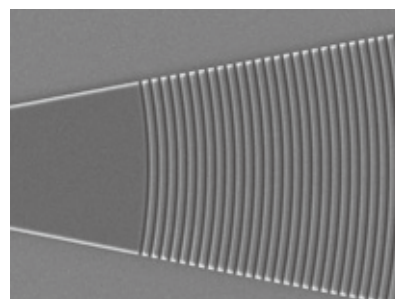
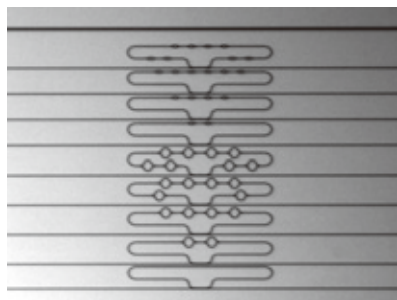
Dot pattern

20 nm gold dots at 100 nm pitch written "on the fly"



Courtesy of M. Rosamond (University of Leeds).

Waveguide



Courtesy of R. Bojko (Nanofabrication Facility, University of Washington).

JBX-3200MV & JBX-3050MV

De-Facto Standard Tools for Wide Range Node Mask Production

Principal Specifications

Writing Method : VSB, Vector Scan, Step & Repeat
Electron Beam (EB) : 50 kV

JBX-3200MV Series

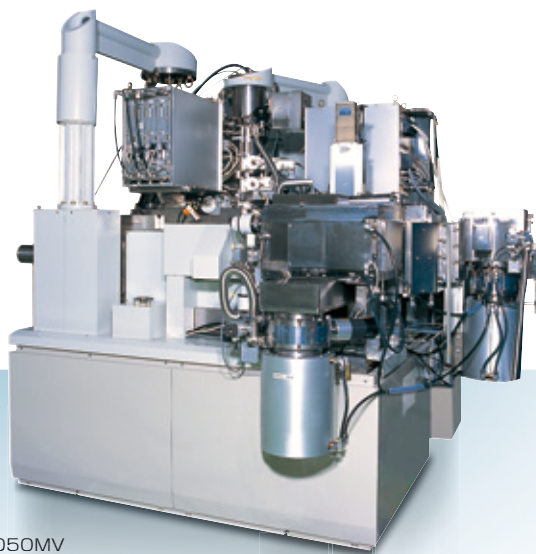
CD uniformity : ≤ 3 nm (3 s)
Image Placement : ≤ 5 nm (3 s)



JBX-3200MV



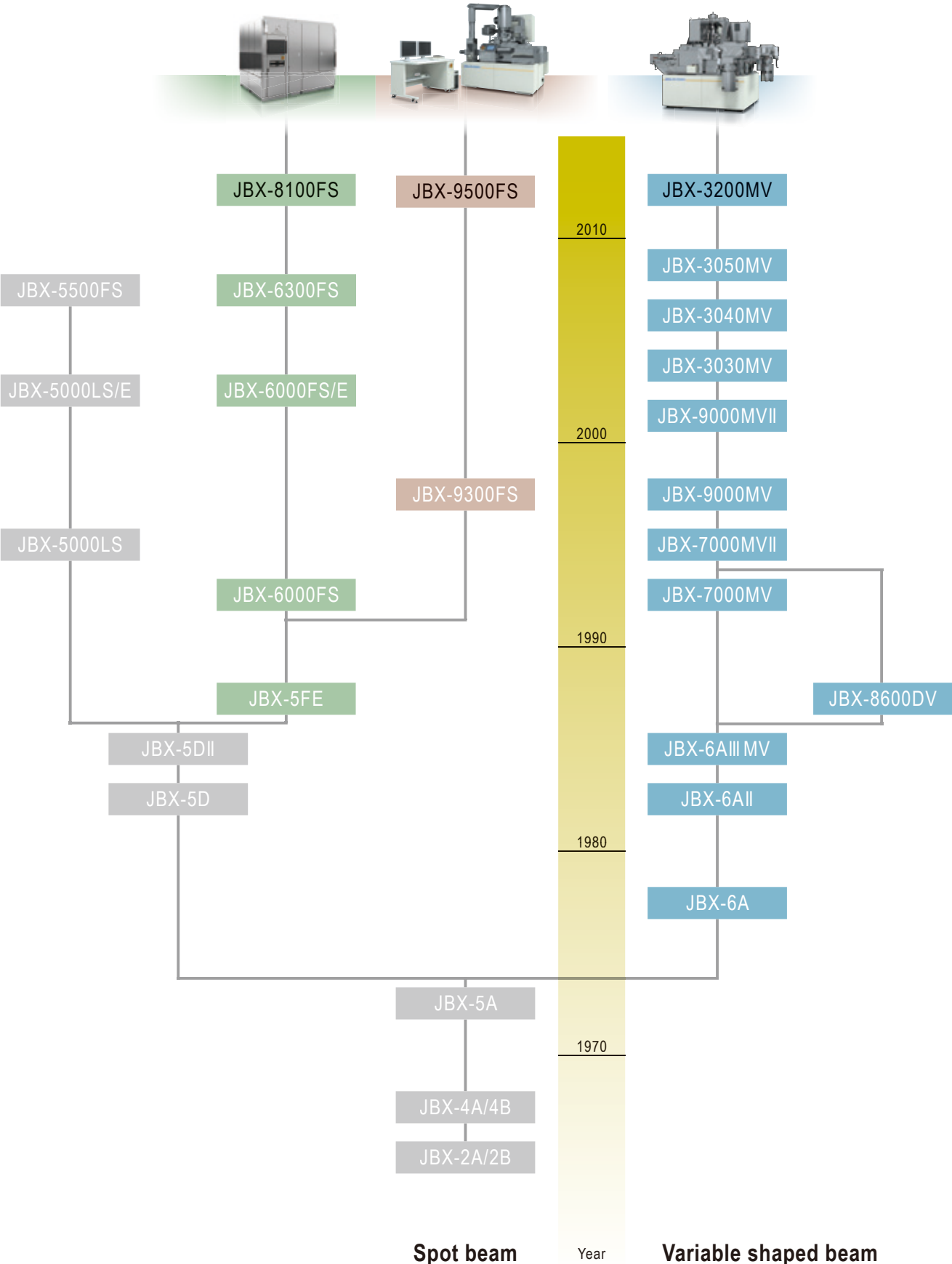
JBX-3050MV



JBX-3050MV Series

CD uniformity : ≤ 3.5 nm (3 s)
Image Placement : ≤ 7 nm (3 s)

Development history of JEOL E-Beam lithography system



*Specifications subject to change without notice.

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